Project for the course "Numerical Integration of Stochastic Differential Equations"

Unbiased estimation via randomization - Application to Option Pricing

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Introduction and probabilistic setting

Let us consider the stochastic basis $(\Omega, \mathcal{F}, \mathbb{P}, (\mathcal{F}_t)_{t\geq 0})$, where the filtration $(\mathcal{F}_t)_{t\geq 0}$ satisfies the usual conditions. Let $Y \in L^2(\Omega)$ be a square integrable random variable. In applications to stochastic differential equations, Y typically has the form

$$Y = \varphi(X_T) \tag{1}$$

where $(X_t)_{t\in[0,T]}$ is the solution of an SDE and $\varphi : \mathbb{R}^d \to \mathbb{R}$ is a Lipschitz measurable function such that $\varphi(X_T) \in L^2(\Omega)$. A frequent goal in this context is to estimate the expectation of Y:

$$\alpha = \mathbb{E}[Y]. \tag{2}$$

It is often the case, however, that one does not have access to Y directly, which represents the exact quantity of interest, but can only simulate suitable approximations of this random variable. For example, the numerical discretization of the SDE gives access to a sequence $(Y_\ell)_{\ell\geq 0} \subset L^2(\Omega)$ that, under suitable assumptions, converges in a suitable sense to Y. In this context, $\ell > 0$ is an accuracy level and is related to the time-step size used to discretize the SDE dynamics. By drawing N i.i.d. replicas of Y_ℓ, a Monte Carlo estimator of (2) is given by

$$\hat{\alpha} = \frac{1}{N}\sum_{k=1}^{N} Y_\ell^{(k)}, \quad Y_\ell^{(k)} \overset{\text{i.i.d.}}{\sim} Y_\ell. \tag{3}$$

The estimator (3) is biased by the time discretization and one can control this bias by decreasing the time-step size. However, to remove the bias entirely, one should let $\ell \to \infty$ thereby incurring infinite computation expenses.

The goal of this project is to exploit multilevel techniques to derive finite variance *unbiased* estimators of (2), which can be simulated with a finite computational budget.

Background: unbiased estimation via randomization

In this section, we introduce the methodology presented in [3] for constructing unbiased estimators of (2). Let $(Y_\ell)_{\ell\geq 0} \subset L^2(\Omega)$ be such that $\|Y_\ell - Y\|_2 \to 0$ for some $Y \in L^2(\Omega)$, and define

$$\Delta_\ell := Y_\ell - Y_{\ell-1}, \qquad Y_{-1} := 0.$$

Let N be a nonnegative integer-valued random variable, almost surely finite, independent of $(Y_\ell)_{\ell\geq 0}$, and such that $\mathbb{P}(N \geq \ell) > 0$ for all $\ell \geq 0$. Define

$$\bar{Z} := \sum_{\ell=0}^{N} \frac{\Delta_\ell}{\mathbb{P}(N \geq \ell)}, \tag{4}$$

the so-called **coupled-sum estimator**.

Theorem 1 in [3] shows that, under the assumption

$$\sum_{\ell=1}^{\infty} \frac{\|Y_{\ell-1} - Y\|_2^2}{\mathbb{P}(N \geq \ell)} < \infty, \tag{5}$$

one has $\mathbb{E}[\bar{Z}] = \mathbb{E}[Y]$, so (4) is an unbiased estimator of the quantity of interest (2). The same theorem also proves that $\bar{Z} \in L^2$ and characterizes its second moment by

$$\mathbb{E}[\bar{Z}^2] = \sum_{\ell=0}^{\infty} \frac{\bar{\nu}_\ell}{\mathbb{P}(N \geq \ell)}, \tag{6}$$

where

$$\bar{\nu}_\ell = \|Y_{\ell-1} - Y\|_2^2 - \|Y_\ell - Y\|_2^2.$$

Therefore, although (4) is unbiased, suitable assumptions on both N and $(Y_\ell)_{\ell \geq 0}$ are needed to ensure that (6) converges and that simulating (4) has finite expected cost.

- **(Q1)**

  Consider the case where $Y = \varphi(X_T)$, with $(X_t)_{t\in[0,T]}$ solution of

  $$\mathrm{d}X_t = f(X_t)\,\mathrm{d}t + g(X_t)\,\mathrm{d}B_t, \qquad X_0 = \xi,$$

  and where $Y_\ell = \varphi(X_T^{(\ell)})$ is obtained from a numerical scheme with time step $h_\ell = T/2^\ell$.

  In this question we compare two different constructions of $(Y_\ell)_{\ell\geq 0}$ and $Y_\infty = Y$:

  – a *Brownian refinement coupling* where all the random variables Y_ℓ are driven by the same Brownian motion,

  – an *independent coupling* where Y_ℓ and $Y_{\ell'}$, for $\ell \neq \ell'$, are generated from independent sources of randomness.

  1. Assume that, for each ℓ, the random variables Y_ℓ and Y are independent, that $\mathbb{E}[Y_\ell] \to \mathbb{E}[Y]$ and $\mathrm{Var}(Y_\ell) \to \mathrm{Var}(Y)$ as $\ell \to \infty$. Show that $\|Y_\ell - Y\|_2 \not\to 0$ in general. Deduce that (5) cannot hold no matter what choice of $\mathbb{P}(N \geq \ell)$, whenever $\mathrm{Var}(Y) > 0$.
  2. Assume that all discretizations are driven by a single Brownian motion $(B_t)_{t\in[0,T]}$ and that, for each $\ell \geq 1$, the pair $(Y_{\ell-1}, Y_\ell)$ is constructed by the *Brownian refinement coupling*, i.e., the Brownian increments used to construct Y_ℓ and $Y_{\ell-1}$ are all consistent with a single Brownian motion path. Assume moreover that the numerical scheme has strong order $p > 0$ under this coupling, i.e.

  $$\|Y_\ell - Y\|_2 \leq C h_\ell^p. \tag{7}$$

  Show that, for a geometric choice $\mathbb{P}(N \geq \ell) = q^\ell$ with $q \in (0,1)$, condition (5) is satisfied provided $q > 2^{-2p}$, and conclude that $\bar{Z}$ has finite variance.

- **(Q2)**

  Let $(\bar{Z}^{(i)}, \bar{\tau}^{(i)})_{i\geq 1}$ be i.i.d. copies of the pair $\big(\bar{Z}, \bar{\tau}\big)$, where

  $$\bar{Z} := \sum_{\ell=0}^{N} \frac{\Delta_\ell}{\mathbb{P}(N \geq \ell)}, \qquad \Delta_\ell := Y_\ell - Y_{\ell-1}, \;\; Y_{-1} := 0,$$

  and $\bar{\tau}$ denotes the (random) computational time required to generate one replicate $\bar{Z}$.

  For a computational budget $c > 0$, define the renewal counting process

  $$\Gamma(c) := \max\left\{ n \geq 0 : \sum_{i=1}^{n} \bar{\tau}^{(i)} \leq c \right\}, \tag{8}$$

  and the time-budgeted estimator

  $$\bar{\alpha}(c) := \begin{cases} \dfrac{1}{\Gamma(c)} \displaystyle\sum_{i=1}^{\Gamma(c)} \bar{Z}^{(i)}, & \Gamma(c) > 0, \\ 0, & \Gamma(c) = 0. \end{cases}$$

  Assume that $\mathbb{E}[\bar{\tau}] < \infty$ and $\mathrm{Var}(\bar{Z}) < \infty$.

  In this setting, the following asymptotic normality result holds (see [3, Eq. (9)]):

  $$c^{1/2}\big(\bar{\alpha}(c) - \mathbb{E}[Y]\big) \implies \big(\mathbb{E}[\bar{\tau}] \cdot \mathrm{Var}(\bar{Z})\big)^{1/2} \mathcal{N}(0,1), \qquad \text{as } c \to \infty. \tag{9}$$

  In this part, we will rely on (9) to find an optimal distribution for N so that $\mathbb{E}[\bar{\tau}] \cdot \mathrm{Var}(\bar{Z})$ is minimized.

1. Express $\mathbb{E}[\bar{\tau}]$ in terms of the cost $\bar{t}_j$, $j \geq 0$, required to simulate each replication of Y_j, $j \geq 0$.
2. For $j \geq 0$, let $\bar{F}_j = \mathbb{P}(N \geq j)$ and define

$$\bar{\beta}_j = \begin{cases} \bar{\nu}_j & \text{if } j \geq 1, \\ \bar{\nu}_0 - \alpha^2 & \text{if } j = 0, \end{cases} \tag{10}$$

with α given by (2). With this definition, we can write $\mathrm{Var}(\bar{Z}) = \sum_{j=0}^{\infty} \frac{\bar{\beta}_j}{\bar{F}_j}$. Consider the minimization problem

$$\min_{\bar{F}=(\bar{F}_j)_{j\geq 0}} \quad \bar{g}(\bar{F}) := \left(\sum_{j=0}^{\infty} \frac{\bar{\beta}_j}{\bar{F}_j} \right) \left(\sum_{j=0}^{\infty} \bar{t}_j \bar{F}_j \right) \tag{11a}$$

$$\text{s.t.} \quad \bar{F}_0 = 1, \tag{11b}$$

$$1 \geq \bar{F}_0 \geq \bar{F}_1 \geq \bar{F}_2 \geq \cdots \geq 0, \tag{11c}$$

$$\sum_{j=0}^{\infty} \bar{t}_j \bar{F}_j < \infty. \tag{11d}$$

Assume that $(\bar{\beta}_j)_{j\geq 0}$ is a nonnegative sequence with $\bar{\beta}_0 > 0$, and that $(\bar{t}_j)_{j\geq 0}$ is a positive sequence. Assume moreover that the sequence defined by

$$\bar{F}_j^{\star} = \frac{\sqrt{\bar{\beta}_j / \bar{t}_j}}{\sqrt{\bar{\beta}_0 / \bar{t}_0}}, \qquad j \geq 0, \tag{12}$$

is feasible for (11). Prove that $\bar{F}^{\star} = (\bar{F}_j^{\star})_{j\geq 0}$ minimizes (11).

Hint: It can be useful to start by proving that $p_i = \frac{\bar{t}_i \bar{F}_i}{\sum_{j=0}^{\infty} \bar{t}_j \bar{F}_j}$, for $i \geq 0$, satisfies the following inequality

$$\left(\sum_{j=0}^{\infty} p_j \sqrt{\frac{\bar{\beta}_j}{\bar{t}_j} \frac{1}{\bar{F}_j}} \right)^2 \leq \sum_{j=0}^{\infty} p_j \left(\sqrt{\frac{\bar{\beta}_j}{\bar{t}_j} \frac{1}{\bar{F}_j}} \right)^2. \tag{13}$$

3. In our setting, it is natural to assume that the cost $\bar{t}_j$ is inversely proportional to the discretization step size h_j. Hence, from now on, we assume that $\bar{t}_j = \mathcal{O}(2^j)$. Assume that the discretization scheme has strong order p, namely

$$\|Y_j - Y\|_2 = \mathcal{O}(2^{-pj}).$$

Show first that

$$\mathbb{E}[\Delta_j^2] = \mathcal{O}(2^{-2pj}), \qquad \Delta_j = Y_j - Y_{j-1}.$$

Then restrict to a geometric sequence

$$\mathbb{P}(N \geq j) = 2^{-rj}.$$

Determine which value of r balances variance and cost, and state for which condition on p this choice gives both $\mathbb{E}[\bar{\tau}] < \infty$ and $\mathrm{Var}(\bar{Z}) < \infty$.
4. Discuss the role of the strong order p of the numerical method. Do you expect that increasing the strong order would significantly improve the overall efficiency?

- **(Q3) Numerical experiments. Option Pricing under Geometric Brownian Motion.**

Modeling asset prices is the standard tool in the evaluation of *options*. An option is a contract between two parties, a "writer" and a "holder". Both sides agree that at a fixed time T, called the *expiry date*, the writer will pay an amount of money to the holder, called the *payoff*, whereas the contract is assumed to be stipulated at time $t = 0$. The value of the *payoff* is determined by the behavior of the price S_t of the chosen asset in the interval of time $[0, T]$. On the other hand, the holder does have to pay a small amount, called "premium", to the writer at time $t = 0$ by contract. For example, let us consider a "European call" option, which is defined by the following payoff function

$$P(S_T) = (S_T - K)^+, \tag{14}$$

where $K > 0$ is the strike parameter and $(\cdot)^+ = \max(\cdot, 0)$. If $S_T > K$, the holder of the option will receive a positive payoff $P(S_T)$ from the writer, otherwise no money will be received. Therefore, option pricing is essentially betting at time $t = 0$ about what will be the value $P(S_T)$, i.e. the random variable S_T, so that the holder and writer can deal the best option for themselves.

Consider the Black & Scholes model for the time evolution of the price of an asset S_t

$$dS_t = \mu S_t \, dt + \sigma S_t \, dB_t, \qquad S_0 = 1, \tag{15}$$

with interest rate $\mu = 0.05$, volatility $\sigma = 0.2$, and $T = 1$. We are interested in estimating the value of a European-Call option written on an underlying asset satisfying (15). The value of this option is given by the expected discounted payoff

$$\mathbb{E}[\varphi(S_T)|S_0], \qquad \varphi(x) = e^{-\mu T} P(x) = e^{-\mu T}(x - K)^+,$$

where $K = 1$ is the strike parameter. Under the Black & Scholes model, the exact value of the option is known to be

$$\mathbb{E}[\varphi(S_T)|S_0 = 1] = 0.104505836. \tag{16}$$

Although S_T can be simulated exactly, you are asked to validate your estimator by using the Milstein time-discretization scheme with mesh $h_\ell = 2^{-\ell}$, with $t_k = kh_\ell$ for $k = 0, \dots, 2^\ell$.

Coupling between levels. In order to construct the level differences $\Delta_\ell = Y_\ell - Y_{\ell-1}$ efficiently, it is standard to simulate $(Y_{\ell-1}, Y_\ell)$ on the *same* probability space using a common Brownian motion $\{B_t\}_{t\geq 0}$ and a Brownian refinement coupling. We detail here the construction for the Euler-Maruyama scheme; for level ℓ, set $\Delta B_k^{(\ell)} := B_{(k+1)h_\ell} - B_{kh_\ell}$, and for level $\ell - 1$ define

$$\Delta B_k^{(\ell-1)} := \Delta B_{2k}^{(\ell)} + \Delta B_{2k+1}^{(\ell)}. \tag{17}$$

Under this coupling, the fine-level approximation $\{S_k^f\}_{k=0}^{2^\ell}$ of (15) via the Euler–Maruyama scheme takes the form

$$\begin{cases} S_{k+1}^f = S_k^f + \mu S_k^f h_\ell + \sigma S_k^f \Delta B_k^{(\ell)}, & \text{for } k = 0, ..., 2^\ell - 1, \\ S_0^f = S_0. \end{cases} \tag{18}$$

The coarse discretization $\{S_k^c\}_{k=0}^{2^{\ell-1}}$ of (15) is given by

$$\begin{cases} S_{k+1}^c = S_k^c + \mu S_k^c h_{\ell-1} + \sigma S_k^c \Delta B_k^{(\ell-1)}, & \text{for } k = 0, ..., 2^{\ell-1} - 1, \\ S_0^c = S_0. \end{cases} \tag{19}$$

This ensures that the coarse and fine discretizations are driven by consistent Brownian increments. This strong coupling is essential for reducing the variance of Δ_ℓ; see [1]. Although (18)–(19) illustrate the construction for Euler–Maruyama, the same Brownian refinement coupling can be used with any time-discretization scheme based on the same increments, in particular with the Milstein scheme, by building both the fine and coarse approximations from the coupled Brownian increments (17).

1. Detail the construction of the Milstein scheme with coupling between levels. In light of your answers to the previous points of the project, justify why, in our context, Milstein is preferable to Euler-Maruyama.
2. **Coupled sum estimator.**

   Implement the coupled-sum unbiased estimator (4) using the Milstein scheme. Assume that the random truncation level N has a geometric tail,

   $$\mathbb{P}(N \geq \ell) = q^\ell.$$

   Choose the parameter q according to the discussion on the approximate optimal distribution developed in the previous section. In particular, using the a priori bounds on $\bar{\beta}_\ell$ and $\bar{t}_\ell$, justify the corresponding choice of q (which, in this setting, leads to the decay $q^\ell = 2^{-3\ell/2}$).

   Consider computational budgets $c = 20000$, 100000, 500000 and generate i.i.d. replicas of $\bar{Z}$ until the cumulative computational cost reaches the prescribed budget. You may assume that the cost of simulating Δ_ℓ is dominated by the fine level and is given by the total number of time steps 2^ℓ. Hence, a computational budget c allows for simulating $\lfloor \frac{c}{2^\ell} \rfloor$ realizations of Δ_ℓ.

Estimate $\mathbb{E}[\varphi(S_T)]$ and briefly discuss the observed performance. In particular, construct an asymptotic 95% confidence interval and report the estimated mean, an estimate of the variance, the total computational cost (e.g. total number of time steps), and the empirical Root-Mean-Squared Error (RMSE) with respect to the reference value (cf. (16)). The RMSE can be estimated by repeating the experiment several times.

3. **Standard Monte Carlo estimator.**
Implement a standard Monte Carlo estimator (3) for $\mathbb{E}[\varphi(S_T)]$ based on a time-discretization of the SDE. Select both the discretization parameter h and the number of samples N in terms of the given computational budget c.
In particular, since the cost of one path is dominated by h^{-1}, then the total budget satisfies

$$Nh^{-1} \approx c.$$

Assuming that the discretization error is of order h^{ζ}, where ζ is the weak order of the numerical method, and the statistical error is of order $N^{-1/2}$, choose h and N so that

$$h^{\zeta} \approx N^{-1/2},$$

thereby balancing discretization and statistical errors. This leads to

$$h \approx c^{-1/(2\zeta+1)}, \qquad N \approx c^{2\zeta/(2\zeta+1)}.$$

For the same computational budget, compare the coupled-sum and standard Monte Carlo estimators in terms of empirical bias, empirical variance, the work–variance product (estimated as average work per replicate times the variance of one replicate), confidence interval width, and overall computational efficiency, as measured by the RMSE as a function of the computational budget. In addition, for each computational budget, construct and plot a 95% confidence interval for the estimated mean, and quantify the empirical bias by comparing the estimator mean with the reference value.

- **(Q4) CIR model.** Let us focus on the CIR process

$$dV_t = \kappa(\theta - V_t)\,dt + \sigma V_t^{1/2}\,dB_t, \quad V_0 = v_0 > 0, \tag{20}$$

where $\kappa, \theta, \sigma > 0$. The CIR process is a standard model for short interest rates and stochastic volatility, since its square-root diffusion coefficient helps preserve positivity. This makes it a natural framework for option-pricing problems in which the underlying source of uncertainty is a positive and mean-reverting financial factor. In fixed-income markets, many derivatives are sensitive to the evolution of interest rates, so short-rate models play a central role in their valuation. In this section, we consider a European call-type payoff written on the terminal value V_T, using the CIR dynamics as a test case.

It is known that (20) admits a unique strong solution V.

1. Prove that V_t is nonnegative for all $t \geq 0$.
Hint: you can consider the following auxiliary process (which also has a unique strong solution)

$$\mathrm{d}V_t = \kappa(\theta - V_t)\,dt + \sigma\sqrt{(V_t \vee 0)}\mathrm{d}B_t^1, \quad V_0 = v_0 > 0, \tag{21}$$

where $(x \vee 0) = \max(x, 0)$, and the stopping time $\tau_\epsilon = \inf\{t \geq 0 \;:\; V_t = -\varepsilon\}$ for any ε small enough. Show by contradiction that it can not happen $\tau_\epsilon < +\infty$.

2. Suppose that $2\kappa\theta > \sigma^2$. Prove that $\mathbb{P}(V_t > 0 \quad \text{for all } t \in [0, T]) = 1$.
Hint: For $\epsilon > 0$, introduce the stopping time $\tau_\epsilon = \inf\{t \geq 0 : V_t \leq \epsilon\}$. Prove that, for each $t \in [0, T]$, $\mathbb{P}(\tau_\epsilon \wedge t < t) \to 0$ as $\epsilon \to 0$. You can start by defining

$$m = \frac{2\kappa\theta - \sigma^2}{\sigma^2}$$

and apply Itô formula to the function $f(x) = x^{-m}$. From there, you can obtain a bound on $\mathbb{E}[V_t^{-m}(\tau_\epsilon \wedge t)]$, and conclude with Chebyshev's inequality.

3. Consider the transformation $Z_t = \sqrt{V_t}$ (called Lamperti transform of (20)). Verify that Z_t satisfies the following SDE
$$dZ_t = \phi(Z_t)\,dt + \frac{1}{2}\sigma\,dB_t, \tag{22}$$
where
$$\phi(x) = \frac{1}{2}\kappa\left(\left(\theta - \frac{\sigma^2}{4\kappa}\right)x^{-1} - x\right).$$
Notice that the Lamperti transform maps (20) into an SDE with additive noise, however with non-Lipschitz drift.

4. One difficulty associated with the numerical integration of (20) is that the approximated solution may not remain positive. Under suitable assumptions, [2] show that drift-implicit schemes preserve the positivity of the numerical solution.

   In this part, we analyze two such numerical schemes for discretizing the dynamics (20). Let us introduce a uniform timestep $h = T/N$ and set $t_k = kh$ and $\Delta B_k = B_{t_{k+1}} - B_{t_k}$ for $k = 0, ..., N$.

   (a) The first scheme discretizes (22) with a drift-implicit Euler-Maruyama scheme, yielding the recursion
   $$Z_{k+1} = Z_k + \phi(Z_{k+1})h + \frac{1}{2}\sigma\Delta B_k, \quad k = 0,\ldots,N-1. \tag{23}$$
   and recovers a numerical approximation of (20) via the inverse transformation
   $$V_{k+1} = Z_{k+1}^2, \quad k = 0,\ldots,N-1. \tag{24}$$

   (b) The second scheme uses a drift-implicit Milstein method directly for (20), given by
   $$\begin{aligned} V_{k+1} = V_k &+ \kappa(\theta - V_{k+1})h + \sigma\sqrt{V_k}\,\Delta B_k \\ &+ \frac{\sigma^2}{4}\big((\Delta B_k)^2 - h\big), \quad k = 0,\ldots,N-1. \end{aligned} \tag{25}$$

   Consider the following parameters
   $$\theta = 0.125, \quad \kappa = 2, \quad \sigma = 0.5, \quad V_0 = 0.125, \quad T = 1. \tag{26}$$

   Discretize the CIR dynamics (20) using (23)-(24) and (25). Plot several sample paths and verify that the schemes preserve the positivity of the numerical approximation.

   Study numerically the strong order of convergence achieved by these numerical schemes. To estimate it, you may use a reference solution obtained by discretizing the CIR dynamics with the same numerical scheme and a very small timestep.

   Finally, what can you say about the order of convergence potentially achieved by the standard (drift-explicit) Euler–Maruyama discretization?

- **(Q5) Numerical experiments: option pricing under the CIR model.**

  Now, we aim to estimate the price of a European call option written under the CIR model. We set the parameters of the models as in (26) and consider the unbiased estimator (4) to estimate the option price $\mathbb{E}[Y]$ with
  $$Y = \varphi(V_T), \quad \varphi(v) = e^{-\mu}(v - K)^+, \tag{27}$$
  with $\mu = 0.05$ and $K = 0.25$.

  Using the discretization scheme (25), perform the following numerical study for the CIR model

  1. implement the coupled-sum estimator (4) based on a geometric distribution for N ($\mathbb{P}(N \geq \ell) = q^\ell$)
  2. implement the standard Monte Carlo estimator (3).

     Consider the computational budgets $c = 1000$, 10000, 50000 and compare all the methods in terms of empirical variance, product between expected work and estimator's variance (estimated as average work per replicate times the variance of one replicate), confidence interval width, and RMSE as a function of computational budget.

References

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